Can Bayes Factors "Prove" the Null Hypothesis?

Michael Smithson, The Australian National University

Email: Michael.Smithson@anu.edu.au

Abstract

It is possible to obtain a large Bayes Factor (BF) favoring the null hypothesis when both the null and alternative hypotheses have low likelihoods, and there are other hypotheses being ignored that are much more strongly supported by the data. As sample sizes become large it becomes increasingly probable that a strong BF favouring a point null against a conventional Bayesian vague alternative co-occurs with a BF favouring various specific alternatives against the null. For any BF threshold q and sample mean $\bar{X}$, there is a value n such that sample sizes larger than n guarantee that although the BF comparing H_0 against a conventional (vague) alternative exceeds q, nevertheless for some range of hypothetical μ, a BF comparing H_0 against μ in that range falls below $1/q$. This paper discusses the conditions under which this conundrum occurs and investigates methods for resolving it.

One attraction of Bayesian statistical methods for researchers is the belief that, unlike the p-value in frequentist null-hypothesis tests, a large Bayes factor (BF) in favour of the null hypothesis and against its alternative is strong evidence for the null hypothesis (Gallistel, 2009; Masson, 2011). While well-informed users of Bayesian statistics may be aware that there is more to it than this, it seems that many users may not be. Unfortunately, especially in large-sample research with small effects, this belief can be outright misleading. With increasing sample size, a substantial BF favouring the null hypothesis against a conventional vague alternative will increasingly co-occur with substantial BFs favouring a range of other alternatives against the null hypothesis. Thus, depending on the choice of alternative, Bayes factors may sharply contradict each other regarding the evidentiary status of the null hypothesis. I provide a guide to the conditions under which relying on the Bayes factor to assess the null can mislead researchers and research consumers, and develop practical suggestions for researchers.

Collectively, Bayesians provide diverse advice on how to deal with point null hypotheses for continuous parameters. Some advocate using the Bayes factor, B_{01}, comparing a null hypothesis H_0 against its complement H_1 as evidence for or against the null (Dienes, 2014; Konijn, et al. 2015; Rouder, et al. 2018; Wagenmakers, 2007). The Jeffreys table (Jeffreys, 1961) linking adjectives such as "substantial" and "strong" with ranges of BFs has been popular in this connection. Other Bayesians do not endorse BFs for continuous parameters[8,9,10] (Gelman, ete al. 2004; Lynch, 2007; Liu & Aitkin, 2008), with some recommending using credible intervals instead for making decisions about the null (Kruschke, 2011; Bolstad & Curran, 2016). Still others have pointed out that the Bayes factor and credible interval provide different criteria for assessing the null (Rouder, et al. 2018), along similar lines to the classic Lindley paper highlighting situations where a strong BF favouring the null co-occurs with a frequentist significance test rejecting it (Lindley, 1957). Those in this last camp recommend not using credible intervals to decide whether or not to reject the null.

Setting aside the use of credible intervals, it is easy to forget that the Bayes factor tells us how more or less likely one hypothesis is than another hypothesis, but does not inform us about the absolute likelihoods of either hypothesis. Thus, it is possible to obtain a large BF when both the null and alternative hypotheses have low likelihoods, and there are other hypotheses being ignored that are much more strongly supported by the data.

We shall see that this is what can happen to a pointwise H_0 and vague H_1 under certain conditions as sample sizes become large. As sample sizes become large it becomes increasingly probable that a strong BF favouring H_0 against a conventional Bayesian vague H_1 co-occurs with a BF favouring various specific alternatives <u>against H_0</u>. Thus, we may simultaneously find strong support for and against the null from the same data. I will call these co-occurrences "Lindley cases", after the so-called Lindley (1957) "paradox".

Lindley cases are most likely to occur when there are small effects and large samples. For a simple example, consider a two-outcome process producing E or ~E on every trial, and let θ denote the proportion of trials yielding E. Let us assume that θ is a binomial variable, setting H_0 is $\theta = 1/2$, H_1 is $\theta \neq 1/2$, and assigning prior probabilities $\pi(\mathrm{H}_0) = \pi(\mathrm{H}_1) = 1/2$. Suppose we observe n trials and k occurrences of E. The likelihood of these data given H_0 is

$$P(k/n \mid \mathrm{H}_0) = \binom{n}{k} 0.5^k 0.5^{n-k}, \tag{1}$$

the likelihood of the data given H_1 is

$$P(k/n \mid \mathrm{H}_1) = \int_0^1 \binom{n}{k} \theta^k (1-\theta)^{n-k}\, d\theta = 1/(n+1), \tag{2}$$

and the Bayes factor for H_0 against H_1 is

$$B_{01} = P(k/n \mid \mathrm{H}_0) \Big/ P(k/n \mid \mathrm{H}_1) = (n+1)\binom{n}{k} 0.5^n. \tag{3}$$

For example, if $n = 985$ and $k = 524$ (so the sample proportion is 0.532) then $B_{01} = 3.344$, which in the Jeffreys table lands in the "substantial" range, 3-10, favouring H_0. However, if we examine Bayes factors, $B(\theta, \theta_0, x)$, comparing certain point-valued alternatives against the null in the neighbourhood from 0.5 to 2*(524/985) – 0.5 = 0.564, we observe that these $B(\theta, \theta_0, x) > 1$. Moreover, for hypothetical $0.51042 < \theta < 0.55349$, $B(\theta, \theta_0, x) > 3$, i.e., a Bayes factor greater than 3 against the null. This is a Lindley case. It occurs because both likelihoods for H_0 and H_1 are relatively small: $P(k/n \mid \mathrm{H}_0)$= 0.003399 and $P(k/n \mid \mathrm{H}_1)$=